MILITARY SPECIFICATION

INDICATOR SET, OIL PRESSURE A/A24J-17

1. SCOPE

1.1 This specification covers the requirements for one type of oil pressure indicator set, designated A/A24J-17.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-675	Coating of Glass Optical Elements (Anti-reflection)
MIL-C-5015	Connector, Electric, AN Type
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-20708	Synchros, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution

FSC 6620

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MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The indicator set shall consist of the following major components:

a. Indicator, Pressure, Synchro EGU-14/A
b. Transmitter, Pressure, Synchro TRU-100/A.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator set except where magnetic materials are essential.

3.4.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator set except where ferrous materials are essential.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.5 Corrosive fumes. Materials, as installed in the indicator set and under the service conditions specified herein, shall not liberate deleterious fumes.

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3.4.6 Protective treatment. When materials are used in the construction of the indicator set that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction

3.5.1 System

3.5.1.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.5.1.2 Reliability. The indicator set components shall have a minimum acceptable mean-time-between failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent.

3.5.1.3 Longevity. The indicator set shall have an operating life span (longevity) of 3,000 hours.

3.5.1.4 Nonstandard parts. When nonstandard electronic parts are used in the construction of the indicator set, nonstandard parts data approval shall be a prerequisite to approval of the indicator set by the procuring activity.

3.5.1.5 Compatibility. When connected as a set, the indicator and transmitter shall operate satisfactorily at room temperature at the test points and within the combined indicator and transmitter scale error and friction tolerances specified herein.

3.5.2 Transmitter

3.5.2.1 The transmitter shall be so designed that it will sense the differential of two pressures and convert the differential measurement into an electrical signal for use with the indicator specified herein. The transmitter shall not require the use of an amplifier to provide this signal to the indicator. The transmitter shall be of simple design and shall be easy to maintain. Bearings, gears, or parts that are subject to frictional wear or brinelling shall be of stainless steel. Elastic constraints only shall be used for guidance and control of all moving parts.

3.5.2.2 Adjustments. The transmitter shall not require any periodic adjustments. However, a means shall be provided for adjusting and correcting indications over a minimum total of 10 percent of full scale which shall be simple to manipulate and of such design that adjustments can be made with a minimum of special tools.

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3.5.2.3 Overpressure stop. An overpressure stop shall be provided to restrain the prime mover within a range of 105 to 130 percent of full-scale deflection.

3.5.2.4 Lubrication. The transmitter shall be so designed that no periodic lubrication will be required during service.

3.5.2.5 Noncorrosive sealing. The electrical system of the transmitter shall be isolated from both the pressure compartments and the atmosphere by a hermetic seal. The pressure sensing element and any part in contact with the pressure medium shall be constructed of materials that will resist the corrosive action of alcohol, aircraft fuels and oils, and hydraulic fluid.

3.5.2.6 Transmitter case. The case dimensions shall be in accordance with figure 1. The case shall be hermetically sealed.

3.5.2.6.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. Other concentrations of filling medium shall not be used except with prior approval of the procuring activity.

3.5.2.7 Gears. When gears are used in the transmitter, they shall be constructed of stainless steel having a minimum Rockwell hardness of 50-C.

3.5.2.8 Power supply. The transmitter shall operate from a 26V 400-Hz single-phase power source having characteristics in accordance with MIL-STD-704.

3.5.2.9 Electrical connectors. Electrical connectors shall be in accordance with MS33678-14S-2P and shall have the polarizing key at the top.

3.5.2.10 Electrical characteristics

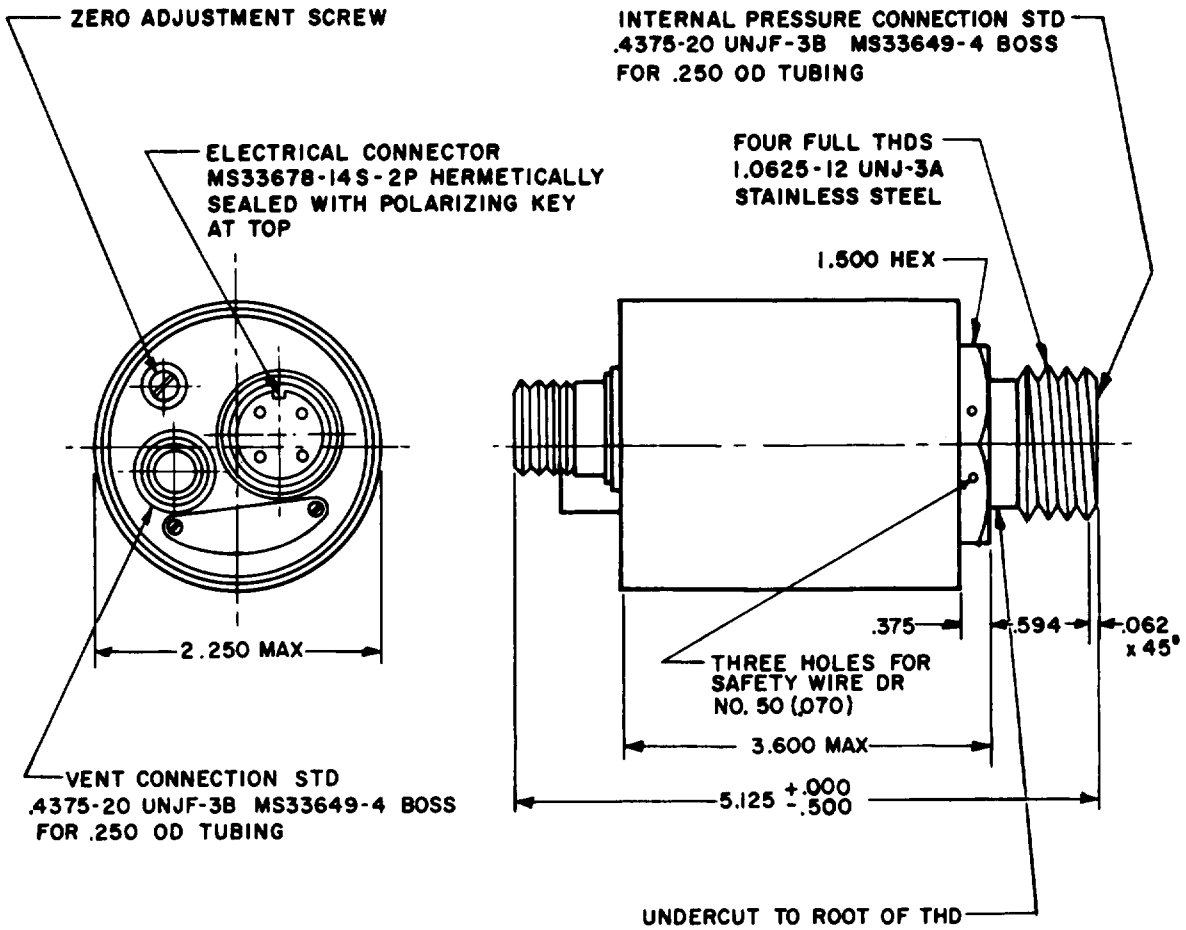
3.5.2.10.1 Synchro. The transmitter synchro shall have the following electrical characteristics with 26V 400-Hz normal excitation applied to the primary (rotor):

3.5.2.10.1.1 Open secondary. The current drawn by the self-synchronous motor shall not exceed 240 ma with secondaries (stator) open circuited.

3.5.2.10.1.2 Power input. Power consumption of the transmitter synchro shall not exceed 6.24 va.

3.5.2.10.1.3 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8 $\pm$0.3V. The variation between the three maximum secondary voltages on the synchro shall not exceed 0.2V.

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ± 0° 15', DECIMALS: ± .016

FIGURE 1. Transmitter Case and Connections

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3.5.2.10.2 Wiring. Transmitter wiring shall be in accordance with figure 2.

3.5.3 Indicator. The indicator shall be designed to provide remote indication of aircraft functions when connected to the transmitter specified herein.

3.5.3.1 Case. The case shall provide a hermetically sealed enclosure for the indicator internal mechanism. It shall be so designed that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by any action to which the indicator may be subjected. The case shall be sufficiently rugged so that small distortions will not affect the accuracy of the indications. The case dimensions shall conform to MS33639 for the 2-inch size. The length shall not exceed 3.5 inches.

3.5.3.1.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. Other concentrations of filling medium shall not be used except with prior approval of the procuring activity.

3.5.3.2 Cap. A separate nonhermetically sealed cap may be used on the back of the case for installing the electrical receptacle. The cap shall be made of nonferrous, low-density metal.

3.5.3.3 Case to cover glass distance. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.5.3.4 Dial. The indicator dial shall conform to figure 3. A minimum diameter of 1-3/4 inches shall be maintained across the outside ends of the graduations. The 30-psi graduation shall be located on the horizontal centerline at the left side of the dial.

3.5.3.4.1 Dial markings. The following markings shall be finished in durable white, color No. 37875 of FED-STD-595. Dimensions of the markings shall be as specified.

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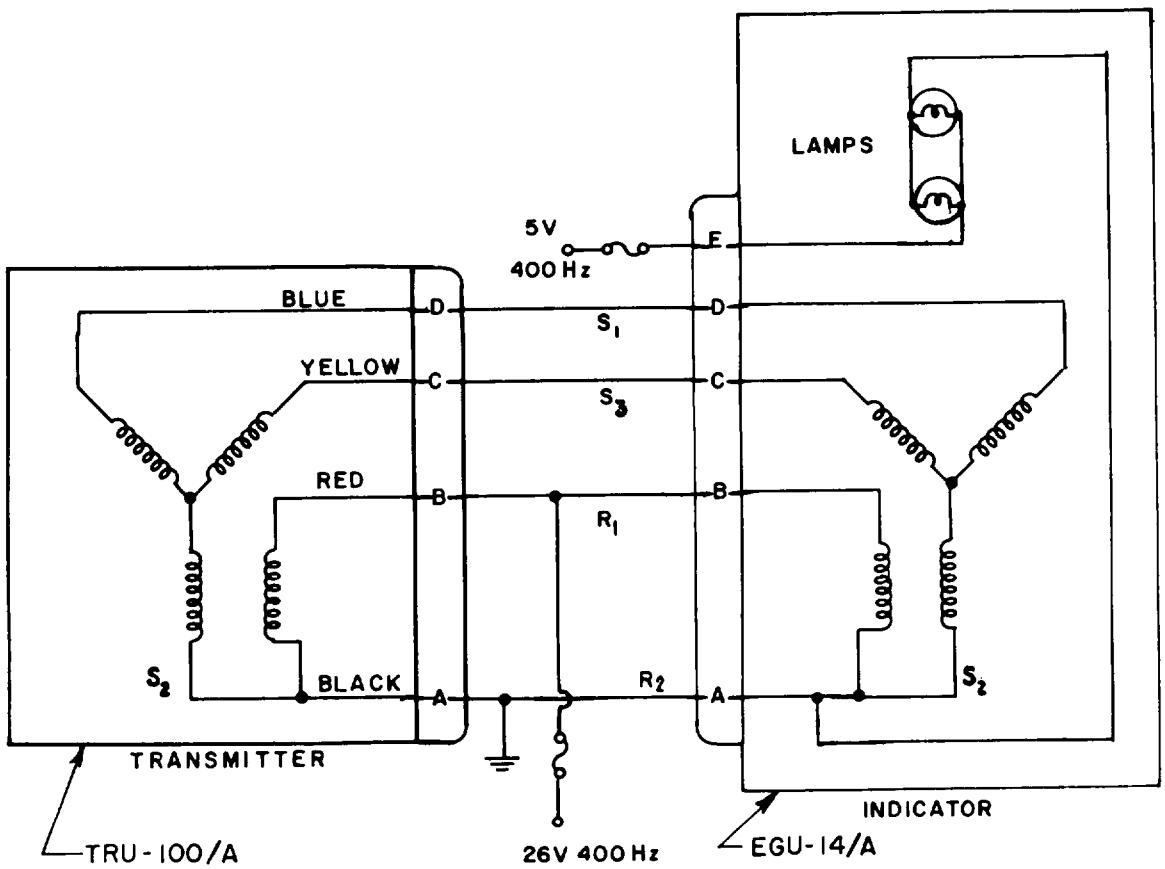
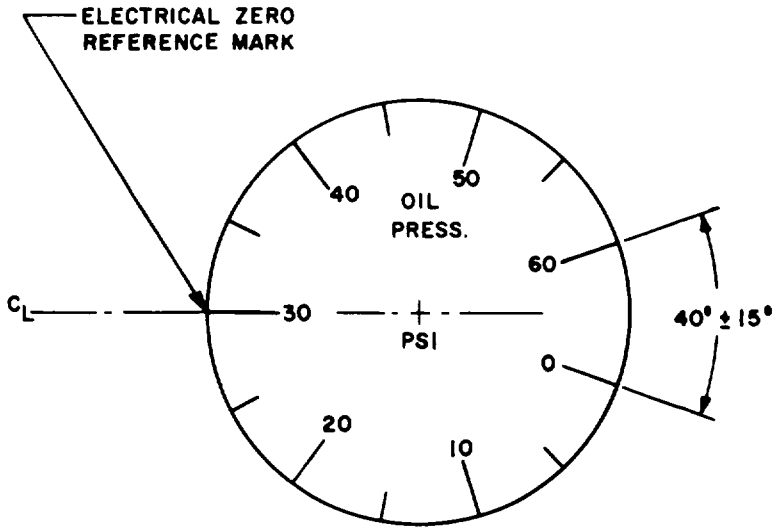
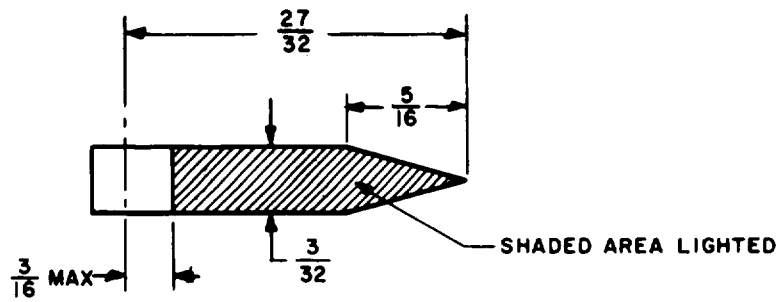


FIGURE 2. Wiring Diagram

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64

FIGURE 3. Dial and Pointer

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Markings not otherwise specified shall be durable black, color No. 37038 of FED-STD-595.

	Height or length Inch ±1/64	Width of line or graduation Inch ±0.005
Numerals 0, 10, 20, 40, 50, and 60	5/32	0.025
10-pound graduations	3/16	0.031
Graduation at 30 psi	9/64	0.031
Lettering OIL PRESS.	1/8	0.020
Lettering PSI	1/8	0.020
5-pound graduations	3/32	0.020

3.5.3.4.2 Dial visibility. Portions of the minor graduations on the dial shall be visible when viewed from any position within the frustum of a cone the side of which intersects the outer edge of the cover glass and which makes an angle of 30° with the perpendicular through the center of the dial.

3.5.3.4.3 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.140 inch.

3.5.3.5 Cover glass. The cover glass shall be clear, flat, and free from flaws that would interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose.

3.5.3.5.1 Percent reflectance. All reflecting glass surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-675, with the exception of the materials used, in addition to withstanding the environmental conditions specified herein. The following reflectance tolerances shall apply:

Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0°	450 - 675	0.6 absolute
0°	425 - 700	0.5 average
30°	400 - 625	1.0 absolute
30°	425 - 700	0.5 average.

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3.5.3.6 Fasteners. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with dial markings or prevent the markings from being located in the proper position.

3.5.3.7 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface. If a tube is used, it shall be of such length as to permit three refillings of the case.

3.5.3.8 Pointer. The pointer shall conform to figure 3 and shall be suitable for integral lighting. The length shall be such that the tip of the pointer will extend into the scale a distance equal to one- to two-thirds the length of the shortest graduation.

3.5.3.8.1 Pointer finish. The lighted portion of the pointer shall be finished in accordance with FED-STD-595, color No. 37875.

3.5.3.9 Lighting. The indicator shall be integrally lighted in accordance with MIL-L-27160.

3.5.3.10 Power supply. The indicator shall operate from a 26V 400-Hz single-phase power source having characteristics in accordance with MIL-STD-704.

3.5.3.11 Electrical connectors. Electrical connectors shall be in accordance with MIL-C-5015. Connector shells shall be protected against corrosion and shall meet the requirements specified herein.

3.5.3.11.1 The indicator shall be provided with an MS33678-14S-5P electrical connector which may be either hermetically sealed or nonhermetically sealed. When the connector is not hermetically sealed, a hermetically sealed baffle plate shall be used to accomplish sealing of the indicator.

3.5.3.12 Electrical wiring. Indicator internal and external wiring shall conform to figure 2. The internal wiring shall be color coded and shall be connected to the receptacle pins as shown on the applicable figure.

3.5.3.13 Electrical characteristics

3.5.3.13.1 Synchro. The indicator synchro shall consist of a salient two-pole wound rotor used as the primary and shall be designed for 26V, 400-cycle, as required, single-phase power, with wye-connected stators as the secondaries. The stators shall be so wound that the amplitudes of the induced voltages measured between any two stator leads will vary approximately sinusoidally with the angular positions of the rotor. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible. The synchro receiver used shall have electrical characteristics compatible with those defined in MIL-S-20708.

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3.5.3.13.1.1 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8V ±3 percent.

3.5.4 Construction. The indicator set shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service.

3.6 Performance

3.6.1 Transmitter. The transmitter shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperatures - operation at temperatures ranging from -54° to +165°C and exposure to temperatures ranging from -62° to +232°C

b. Vibration - vibration of 20g from 5 to 2,000 Hz

c. Temperature-altitude - pressure equivalent to a 100,000-foot altitude at -55°C

d. Salt fog - exposure to salt sea atmosphere for 48 hours

3.6.1.1 Tolerances. The transmitter shall operate within the tolerances specified in tables I and II.

3.6.1.2 Damping. The time required for the change from 60 psi (full-scale pointer deflection) to 25 psi shall not exceed 2 ±1 seconds under the conditions specified in section 4.

3.6.1.3 Vent boss and connector strength. The vent boss and connector shall withstand 25 foot-pounds bending moment applied to the vent boss and connector without evidence of calibration shifting out of tolerance, distortion of the base plate, or damage to welds and soldering.

3.6.1.4 Overpressure. The transmitter shall withstand a 100-percent overpressure for a period of 10 minutes without exceeding the scale error tolerance specified in table I.

3.6.1.5 Case leakage. The case leakage shall not exceed 0.1 micron cubic foot per hour.

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TABLE I. Scale Error Tolerances

Test Pressure PSI	Tolerances ±PSI				
	Room Temperature	Overpressure Power Supply	Low Temperature High Temperature Low-Temperature Exposure High-Temperature Exposure Vibration Endurance Room Temperature after 165° C Exposure	232°C Extreme High Temperature	Sensors Matching Full Range (60 psi)
0	1.0	1.5	2.0	2.5	0
10	1.0	1.5	2.0	2.5	2.5
20	1.0	1.5	2.0	2.5	2.5
30	1.0	1.5	2.0	2.5	2.75
40	1.0	1.5	2.0	2.5	3.0
50	1.0	1.5	2.0	2.5	3.25
60	1.0	1.5	2.0	2.5	3.5

TABLE II. Pointer Error and Friction Tolerances

Tests	Tolerance ±PSI
Friction:	
Room temperature	0.5
High temperature	1.0
Low temperature	1.0
Endurance	1.0
Position error at 40 psi	0.5
Vibration:	
Pointer oscillation	1.0
Pointer variation	1.0

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3.6.1.6 Dielectric strength. The transmitter shall withstand application of 200V rms at a commercial frequency for a period of 5 minutes.

3.6.1.7 Power supply variations. The transmitter shall meet the power supply variations of MIL-STD-704 within the tolerances specified in table I.

3.6.1.8 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 for class ID equipment.

3.6.1.9 Endurance. The transmitter shall withstand 60,000 applications of cycling from zero indication to 80 ±5 percent and back to zero at a rate of 30 ±10 cpm.

3.6.2 Indicator. The indicator shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperature - operation at temperatures ranging from -55 ±2°C to 71°C and exposure to temperatures ranging from -65° to +71° C

b. Vibration - vibration of 2g's from 5 to 500 Hz

c. Temperature altitude - pressure equivalent to a 100,000-foot altitude at -54° C

d. Thermal shock - eight cycles of alternate immersions in tap water at 85° ±2° C and 5° ±2° C

e. Salt fog - exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration

3.6.2.1 Tolerances. The indicator shall operate within the tolerances specified in tables III and IV.

TABLE III. Temperature Scale Error

Test Pressure PSI	Scale Error	
	Room Temperature PSI	High and Low Temp Exposure High and Low Temperature PSI
0	1.0	1.5
10	1.0	1.5
20	1.0	1.75
30	1.0	1.75
40	1.0	1.75
50	1.0	1.75
60	1.0	1.75

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TABLE IV. Position, Friction, and Vibration Error

Test	Tolerance ±PSI
Position error at 30 psi	1.0
Friction	
Room temperature	1.0
High temperature	1.0
Low temperature	1.5
Vibration	
Pointer oscillation	1.0

3.6.2.2 Dielectric strength. The indicator shall withstand 200V rms at a commercial frequency for a period of 5 seconds.

3.6.2.3 Case leakage. Case leakage shall not exceed 0.1 micron cubic foot per hour.

3.6.2.4 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ±5°C for a minimum period of 2 hours and then the face of the cover glass immediately immersed in an ice bath for 3 minutes.

3.6.2.5 Effect of lead length. When the lead length between the indicator and transmitter is changed from 5 to 500 feet, the indication shall not change more than 1° of arc. The ground lead shall be kept short.

3.6.2.6 Magnetic effect. The compass deflection shall not exceed 3° of arc when the indicator is positioned 5-1/2 inches from the compass and the horizontal magnetic field intensity at the indicator is 0.17 to 0.19 oersted.

3.6.2.7 Unit torque gradient. For synchros, the unit torque gradient shall be a minimum of 2,000 milligram-millimeters per angular degree of rotation.

3.6.2.8 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for class ID equipment.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part numbers shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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3.8 Finishes and protective coatings

3.8.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, and case inserts need not be anodized.

3.9 Weight

3.9.1 Transmitter. The weight of the transmitter shall not exceed 1.5 pounds.

3.9.2 Indicator. The weight of the indicator shall not exceed 0.60 pound.

3.10 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11.1 Nameplate. A nameplate shall be permanently attached to the rear of the case. Where applicable, the information marked thereon shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metal decals be used.

3.12 Workmanship. The indicator set shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.12.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability or performance of the indicator set, they shall be held or limited accordingly.

3.12.2 Cleaning. The indicator set shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, or other foreign material removed during and after final assembly. Burrs, sharp edges, and resin flash that might crumble shall also be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicator sets shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Wherever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Transmitter

4.3.2.1 Tapping. Unless otherwise specified, the transmitter shall be lightly tapped or vibrated before each reading.

4.3.2.2 Test voltage. Unless otherwise specified, the transmitter shall be tested using the voltage and frequency specified in 3.5.2.8.

4.3.2.3 Test standard. The transmitter shall be tested with an indicating standard consisting of a standard production indicating test unit calibrated to a maximum error of $\pm 1/4$ percent with a standard ratio transformer circuit as shown on figure 1 or ± 0.5 of a circular degree against a primary transmitting master.

4.3.3 Indicator

4.3.3.1 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.3.3.2 Electrical power. Unless otherwise specified the indicator shall be tested with 26 $\pm$IV, 400 $\pm$5-Hz single-phase a-c power and with 4.5 $\pm$0.5V ac or dc applied to the lighting circuit.

4.3.3.3 Tapping. Unless otherwise specified, the indicator shall be lightly tapped or vibrated before each reading.

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4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of eight indicators and eight transmitters representative of the production equipment. Three of the samples shall be subjected to all the tests except reliability and the other five samples shall be subjected to reliability only. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each transmitter and each indicator shall be subjected to the following tests, as applicable, as described under 4.6:

a. Transmitter

(1) Examination of product

(2) Dielectric strength

(3) Sensor matching

(4) Scale and friction error

(5) Position error

(6) Sealing.

b. Indicator

(1) Examination of product

(2) Individual tests of MIL-L-27160

(3) Dielectric strength

(4) Scale and friction error

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(5) Position error

(6) Sealing.

4.5.2 Sampling tests. One transmitter and one indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests, as applicable, as described under 4.6:

a. Transmitter

(1) Individual tests

(2) Vibration

(3) Low temperature

(4) High temperature

b. Indicator

(1) Individual tests

(2) Sampling plan A tests of MIL-L-27160

(3) Vibration

(4) Low temperature

(5) High temperature

(6) Fogging.

4.5.2.1 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of the failure have been determined and appropriately corrected.

4.5.2.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.6 Test methods

4.6.1 Transmitter

4.6.1.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.1.2 Dielectric strength. A potential of 200V at a commercial frequency shall be applied between any pin of the electrical receptacle and the transmitter case for a minimum of 5 seconds. There shall be no insulation breakdown.

4.6.1.3 Sensor matching. The transmitter shall be subjected to the following sensor matching test:

a. Relative to the atmosphere, a pressure equal to 10 psi of the transmitter range shall be applied to the P_2 (low pressure) port of the transmitter. A pressure shall be applied to the P_1 (high pressure) port which shall exceed that applied to the P_2 port by each of the test points listed in table I except any test point which requires that a pressure be applied to the P_1 port the value of which, relative to the atmosphere, exceeds 120 psi. This test shall be repeated using 20, 30, 40, and 50 psi in lieu of 10 psi.

b. The P_1 and P_2 ports shall be connected together and one pressure applied to the two ports simultaneously. The value of this pressure, relative to the atmosphere, shall be slowly varied continuously from zero through 120 psi and back to zero. The indication shall not differ from zero at any time by more than the value called out in table I, column entitled "Sensors Matching Full Range (60 psi)".

c. A vacuum of 14.0 $\pm$0.5 psi shall be applied to the P_1 port which shall exceed that applied to the P_2 port by each of the test points and a scale error and friction test performed. The scale error shall not exceed the value specified in table I.

4.6.1.4 Scale and friction error. The transmitter shall be tested for scale error at room temperature at the test points specified in table I. The transmitter shall be lightly tapped or vibrated before each reading to remove normal friction. The scale errors shall not exceed the tolerances specified in table I. The difference between the readings before and after the transmitter is tapped or vibrated shall constitute friction error. The friction error shall not exceed the tolerances specified in table II. This test shall be conducted both up-scale and down-scale at the specified test points.

4.6.1.5 Position error. The transmitter shall be energized to obtain a reading of approximately 1/2 scale. The change in indication produced by rotating the transmitter from the normal operating position to a position 90° cw and ccw (longitudinal axis vertical) shall be noted. The maximum change in indication shall not exceed the tolerance specified for position error in table II.

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4.6.1.6 Sealing. The transmitter shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.6.1.7 Vibration. The transmitter shall be subjected to a vibration test in accordance with MIL-STD-810, test method 514, equipment category (b), curve L. Test conditions shall be applied to the transmitter to provide an indication of approximately midscale. Pointer oscillation and variation shall not exceed the tolerance specified in table II. At the conclusion of this test, the transmitter shall be subjected to the scale error test. The scale error shall not exceed the tolerances specified in table I. The transmitter shall not be damaged as a result of this test.

4.6.1.8 Low temperature. The low temperature test shall be conducted in accordance with MIL-STD-810, method 502, procedure I. The transmitter shall be subjected to a temperature of $-54^{\circ} \pm 2^{\circ}$C for a period of 4 hours. At the end of this period and while still at the low temperature, the transmitter shall be subjected to the scale and friction error test. Scale and friction errors shall not exceed those specified in tables I and II.

4.6.1.9 High temperature. The transmitter shall be subjected to a high temperature test in accordance with MIL-STD-810, method 501, procedure I, at temperature of $165^{\circ} \pm 2^{\circ}$C for a period of 6 hours. During this test, the transmitter shall be electrically connected to an appropriate indicator with power applied. At the end of the 6-hour period and while still at the high temperature, the transmitter shall be subjected to the scale and friction error test. The scale and friction error shall not exceed the tolerances specified in tables I and II.

4.6.1.10 Electrical characteristics. The electrical characteristics of the transmitter shall be tested to determine compliance with the requirements specified under 3.5.2.10.

4.6.1.11 Low temperature exposure. The transmitter shall be subjected to a low temperature exposure test in accordance with MIL-STD-810, method 502, procedure I, at a temperature of $-62^{\circ} \pm 2^{\circ}$ C for a minimum of 48 hours. At the end of this period and while still at the specified temperature, the transmitter shall be operated throughout its range. The temperature shall then be increased to $-54^{\circ} \pm 2^{\circ}$ C and the transmitter exposed for an additional 24 hours. The output of the transmitter shall be within the accuracy tolerances specified in tables I and II for low temperature. The transmitter shall then be returned to room temperature for a minimum of 4 hours after which it shall be subjected to the scale error test at room temperature. The scale error shall not exceed the tolerance specified in table I for room temperature.

4.6.1.12 High temperature exposure. The transmitter shall be placed in a chamber and subjected to a temperature of $230^{\circ} \pm 10^{\circ}$ C for a period of 80 hours. While at the high temperature, the transmitter shall be subjected to the scale error test and shall meet the tolerances specified in table I for high temperature exposure. The transmitter shall then be returned to room temperature for a minimum of 4 hours after which it shall again be subjected to the scale error test. The scale error shall not exceed the tolerance specified in table I for room temperature and the transmitter shall not be damaged as a result of this test.

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4.6.1.13 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, method 504, procedure I, equipment class 3 (except the test shall be conducted at -54°C only) with the chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change shall not exceed an average of 3 psi per second while under these conditions. At the end of a 1-hour period, the scale and friction error test shall be conducted. The errors shall not exceed the tolerances specified in tables I and II. This test may be conducted concurrently with the low temperature exposure test.

4.6.1.14 Endurance. The transmitter shall be operated from zero indication to 80 ±5 percent of full scale and back to zero for 60,000 applications at a rate of 30 ±10 cpm. Not less than 1 hour after this test, the transmitter shall be tested for scale error. The scale error shall not exceed the tolerance specified in table for endurance. The transmitter shall be examined to determine that no undue wear has occurred.

4.6.1.15 Electromagnetic interference. The transmitter shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462, for class ID equipment, to 400 megacycles.

4.6.1.16 Power supply variations. The transmitter shall meet the scale error and friction tolerances specified herein when subjected to power supply variations as specified in MIL-STD-704.

4.6.1.17 Salt fog. The transmitter shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I, for a period of 50 hours. The transmitter shall be examined and there shall be no evidence of external corrosion or deterioration that would affect subsequent operation. The external surface of the transmitter may be washed with distilled water and air-dried after which the transmitter shall be subjected to the dielectric strength test. The measured resistance shall be a minimum of 20 megohms.

4.6.1.18 Overpressure. The transmitter shall be subjected to 100 percent overpressure for a period of 10 minutes. At least 1 hour after conclusion of this test, the transmitter shall be subjected to the scale error test using increasing pressure only. The scale error shall not exceed the tolerances specified in table I.

4.6.1.18.1 One of the test samples shall be subjected to 300 psi for 10 minutes and then immersed in water to demonstrate that the internal mechanism has not burst to the extent that it will leak air. The hermetic seal shall be punctured to determine if the internal pressure compartments have ruptured. The transmitter need not be checked for accuracy after this test. The detection of leakage shall be cause for rejection.

4.6.1.19 Damping. A pressure sufficient to produce full-scale pointer deflection shall be applied to the transmitter and released by a quick-release valve. The time required for the change from 60 psi (full-scale deflection) to 25 psi shall be 2 ±1 seconds. The lubricating oil used for this test shall conform to MIL-L-7808 and shall have a viscosity of approximately 2,500 centistokes, such as grade 1005 oil at -54°C.

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4.6.1.20 Vent boss and connector strength. The vent boss and connector mounting shall be sufficiently rigid to withstand a force of 25 foot-pounds bending moment applied in any direction perpendicular to the longitudinal axis of the transmitter without evidence of calibration shifting-out-of tolerance, distortion of base plate, or damage to vent boss and connector welds or soldering. This test shall be conducted with the transmitter secured by the engine mount connection or by the adjacent hex nut only. The transmitter shall then be subjected to the sealing test at room temperature. The scale error shall not exceed the tolerance for room temperature specified in table I.

4.6.2 Indicator

4.6.2.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and markings. Prior to assembly, dials, pointers, and synchros shall be inspected as to markings, dimensions, and finishes.

4.6.2.2 Dielectric strength. A potential of 200 V at a commercial frequency shall be applied between any pin of the electrical receptacle and the case of the indicator for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.6.2.3 Scale and friction error. The indicator shall be tested for scale error at room temperature at the test points specified in table III. The indicator shall be lightly tapped or vibrated before each reading to remove normal instrument friction. The scale errors shall not exceed the tolerances specified in table III. The difference between the readings before and after the indicator is tapped or vibrated shall constitute friction error. The friction error shall not exceed the tolerances specified in table IV. This test shall be accomplished both up-scale and down-scale at the test points specified.

4.6.2.4 Position error. The indicator shall be positioned at the electrical zero reference mark with power applied. The change in pointer position when the indicator is held in any position shall not exceed the position error tolerance specified if table IV.

4.6.2.5 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.6.2.6 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category (a), curve B. A synchro signal shall be maintained on the stators of the indicator synchro to provide an indication of 45 psi. Pointer oscillation shall not exceed the tolerance specified in table IV. Upon completion of this test, the indicator shall be subjected to the scale error test at room temperature. The scale errors shall not exceed the tolerance specified in table III for room temperature.

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4.6.2.7 Low temperature. The indicator shall be subjected to a temperature of -55° ±2°C for a period of 4 hours. While at this temperature, it shall be checked for scale and friction errors and shall not exceed those specified in table III and IV. In every case, the indicator shall not require more than 3 seconds to come within the tolerance specified.

4.6.2.8 High temperature. The indicator shall be subjected to a temperature of 71°C for a period of 4 hours. While at this temperature, it shall be checked for scale errors. The scale error and friction tolerances shall not exceed the values specified in tables III and IV. The response time shall not exceed 3 seconds.

4.6.2.9 Fogging. The indicator shall be placed in a controlled ambient temperature of 85° ±5°C for a minimum period of 2 hours. Within 10 seconds after removal from the oven, the indicator shall be immersed to a depth of 1/4 inch in an ice water bath maintained at 0° to 7°C. After 3 minutes, the indicator shall be removed from the fluid bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected and any evidence of fogging or damage shall be cause for rejection.

4.6.2.10 Secondary voltage. The maximum open-circuit voltage shall be measured between any two secondary leads and shall be 11.8V ±3 percent. The variation between the three maximum secondary voltages on the indicator synchro as measured between any two stator leads shall not exceed 0.2V.

4.6.2.11 Magnetic effect. The indicator, first, not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 3°.

4.6.2.12 Effect of lead length. The indicator shall not change more than 1° of arc when the lead length between the indicator and transmitter is changed from 5 to 500 feet of No. 20 AWG gage wire. Equivalent resistances may be substituted for the added wire length. The ground lead shall be kept short.

4.6.2.13 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, method 504, procedure II (except the test shall be conducted at -54°C only) with a chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change shall not exceed an average of 3 psi per second while under these conditions and at the end of a 1-hour period, the scale and friction error test shall be conducted. The errors shall not exceed the tolerances specified in tables III and IV. This test may be conducted concurrently with the low temperature exposure test.

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4.6.2.14 Low temperature exposure. The indicator shall be subjected to a temperature of -65° ±2°C for a period of 48 hours. While at this temperature, the indicator shall be operated to assure that the pointer moves over the entire scale range without exhibiting sticky or erratic characteristics. The temperature shall then be raised to -54° ±2°C and held for a period of 24 hours. While at this temperature, the indicator shall be subjected to the scale and friction error test, except that in every case, the indicator shall not require more than 4 seconds to come within the tolerances of table IV. The scale and friction error shall not exceed the tolerances specified in tables III and IV. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator again subjected to the scale and friction error test. The scale errors shall not exceed the tolerance specified in table III. There shall be no damage or deterioration that would affect subsequent operation.

4.6.2.15 High temperature exposure. The indicator shall be subjected to a temperature of 71°C for a period of 24 hours and then tested at this temperature for scale errors as specified in 4.6.2.3. The scale and friction error shall not exceed the tolerances specified in tables III and IV. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator subjected to the scale error and friction test. The scale errors observed shall not exceed the tolerance specified in table III. There shall be no damage or deterioration that might affect subsequent operation.

4.6.2.16 Thermal shock. The indicator shall be subjected to eight cycles of immersion in water at a temperature of 85° ±2°C and then in water at a temperature of 5° ±2°C without evidence of moisture penetration or damage to the coating or the case. The length of time for each immersion shall be 30 minutes, with not more than 5 seconds between immersions. After eight continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.6.2.5.

4.6.2.17 Salt fog. The indicator shall be subjected to a salt fog for a period of 48 hours in accordance with MIL-STD-810, method 509, procedure I. There shall be no evidence of external corrosion or deterioration that might affect subsequent operation. The scale errors conducted after this test shall not exceed the tolerances specified for room temperature scale error.

4.6.2.18 Electromagnetic interference. The indicator shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462, for class ID equipment, to 400 megacycles.

4.6.2.19 Unit torque gradient. The indicator shall be connected to the master transmitter synchro and the master transmitter locked in the electrical zero position. The synchro rotors shall then be energized with rated voltage and frequency +1 percent. Weights shall be applied to a torque pulley which is positioned on the rotor shaft to rotate the shaft in a ccw direction through an angle of 5° of arc. The torque tending to return the shaft to its original position shall be computed and shall be no less than 2,000 milligram-millimeters per degree of displacement.

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4.6.2.20 Integral lighting. The integral lighting tests shall be conducted in accordance with MIL-L-27160 in the sequence specified therein.

4.6.3 Reliability. Five transmitters and five indicators shall be subjected to a reliability test. Acceptance permitting production and the total test time shall be in accordance with MIL-STD-781, test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781.

4.6.3.1 Test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, input pressure or signals shall be applied to the transmitter or indicator at a rate of 30 $\pm$5 cpm from 20 to 50 psi of full scale and return. When the transmitter and indicator are tested as a set, they shall be electrically connected and the appropriate power supply furnished during all operating periods with the same cycling as specified above. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator and/or transmitter shall be vibrated in accordance with test level J of MIL-STD-781. The temperature cycles shall be conducted in the following sequence: 4 hours at -20°C, 14 minutes for temperature change to +50°C at a minimum rate of 5°C per minute; 4 hours at +50°C, 14 minutes for temperature change to -20°C, and 4 hours at -20°C. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. Upon completion of each 250 hours of cycling, the indicator and/or transmitter shall be checked for scale error. The scale error reading shall not exceed 150 percent of the room temperature tolerance in the applicable table. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-781 which shall contain test procedures, details of any malfunctions, and corrective actions taken.

4.6.3.2 Longevity. At the completion of the reliability test, testing of three of the indicators and three of the transmitters shall be continued for a total of 3,000 hours each. The test time accumulated during the reliability test shall be applied to this test. Accept-reject criteria is not applicable.

4.6.4 Compatibility. The indicator and transmitter shall be connected as a set and operated at room temperature at the test points shown in tables I and III. The system shall operate within the combined indicator and transmitter tolerances for scale error and friction specified in tables I through IV.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

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6. NOTES

6.1 Intended use. The A/A24J-17 indicator set covered by this specification is intended for use in indicating remotely the oil pressure in aircraft systems that require integrally lighted instruments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Levels of packaging and packing required.

Custodian:
Air Force

Preparing activity:
Air Force - 11

Review activity:
Air Force - 71

Project No. 6620-F041

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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